Quantifying Attention Flow in Transformers

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Transformer Models

Basics

Proposed by famous paper *Attention Is All You Need* [2].

- Numerous applications in NLP, Computer Vision, ...
- Process sequence of input **tokens**
- Stack of identical layers
- Each applying **Self-Attention**

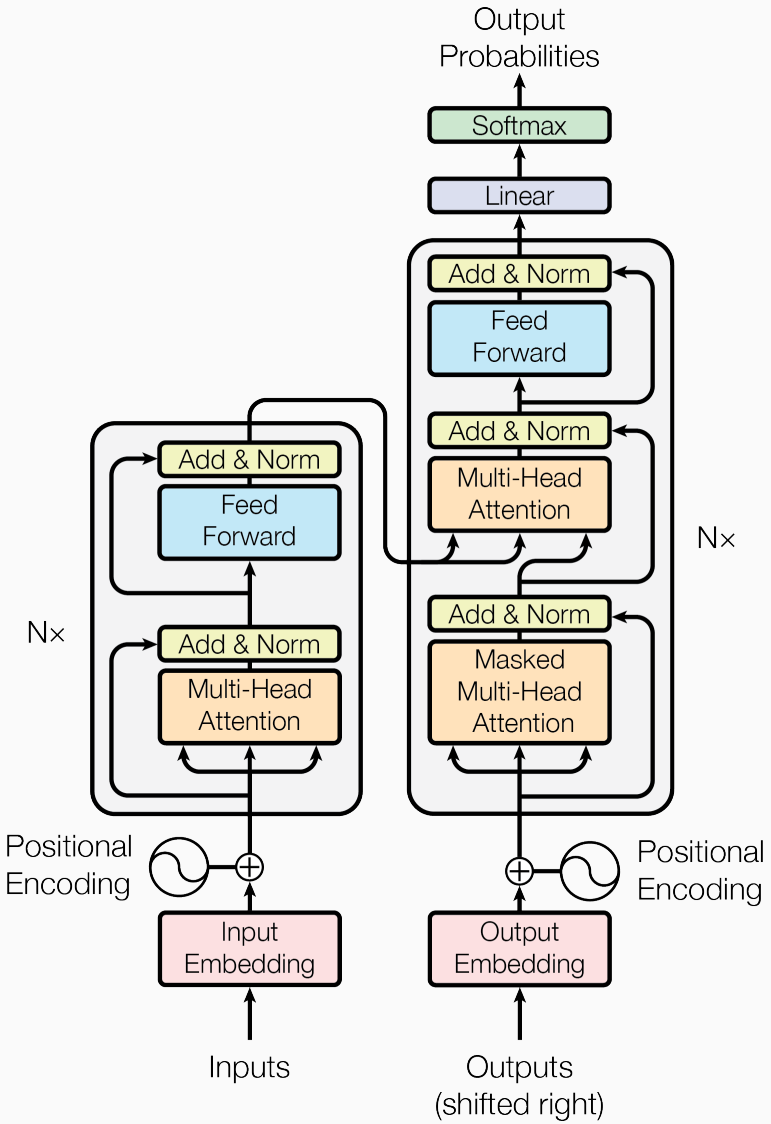

Figure 1: Transformer Model. Taken from [2].

Self-Attention

- For sequence of input tokens $X = (x_1, \ldots, x_n) \in \mathbb{R}^{n \times d}$ compute self-attention with:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V$$

- where the matrices Q, K, and V are different linear projections of the input tokens

Result

After this operation, each token captures **contextual information from all other tokens**.

Problem Statement

Token Processing

- Self-attention operation is executed in each layer
- Across layers, information from different tokens gets increasingly mixed

Research Question: *How can we quantify this information flow across multiple layers?*

Idea

Model information flow in the network with a **Directed Acyclic Graph (DAG)**:

- **Nodes**: Input tokens & hidden embeddings
- **Edges**: Attentions from nodes in each layer to those in previous layer

Verb Number Prediction Task

Predicting singularity or plurality of a verb of a sentence, when the input is the sentence up to the verb position

Attention Flow DAG

Input: *the key to the cabinets . . .* [**is** / **are**?]

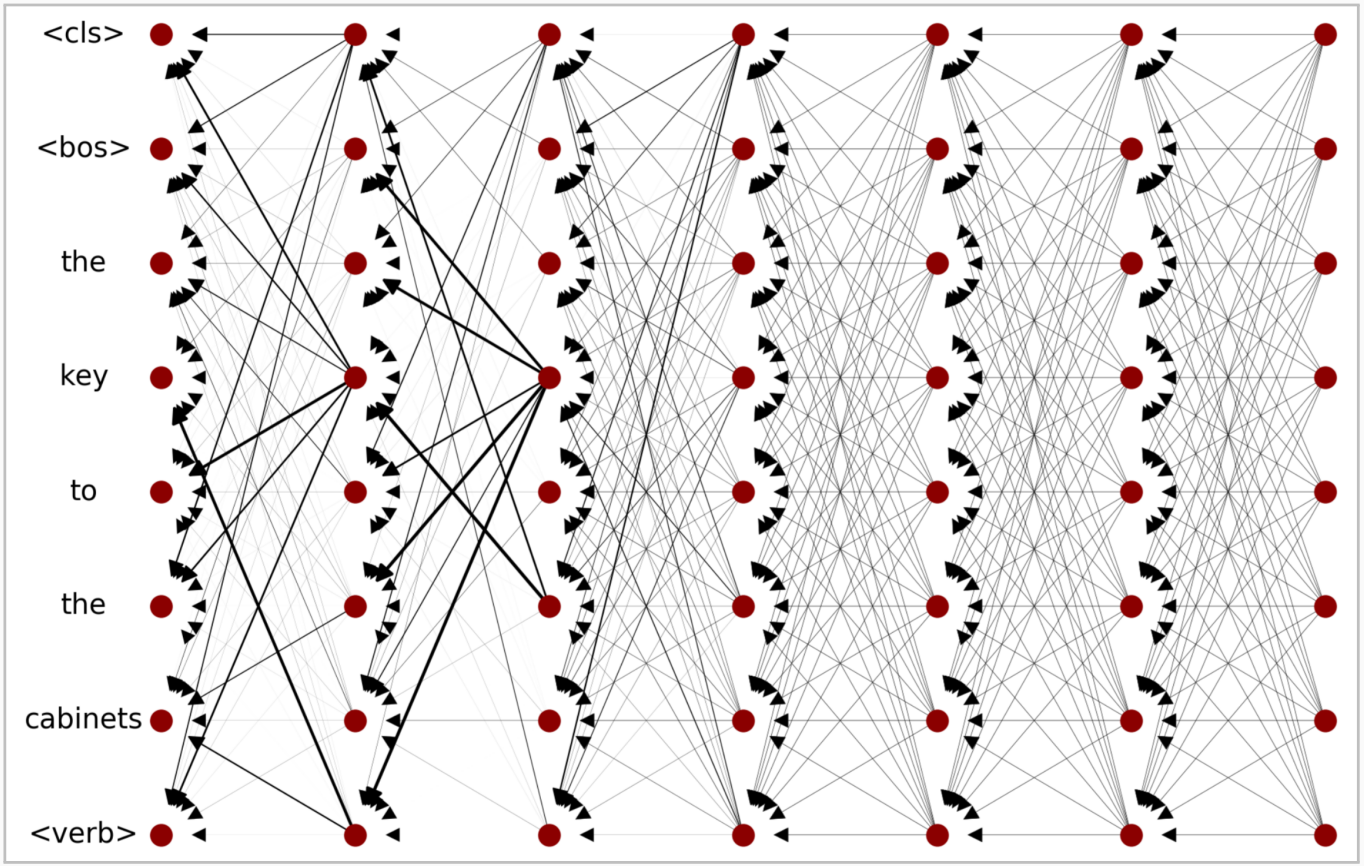

Figure 2: Attention Flow Graph of tokenized input sentence.

References

[1] S. Abnar and W. Zuidema, **Quantifying attention flow in transformers,** 2020. arXiv: 2005.00928 [cs.LG]. [Online]. Available: https://arxiv.org/abs/2005.00928.

[2] A. Vaswani et al., **Attention is all you need,** 2023. arXiv: 1706.03762 [cs.CL]. [Online]. Available: https://arxiv.org/abs/1706.03762.